

Load Testing & Bottlenecks

Software Architecture

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Aside

Github Classroom links for this practical can be found on Edstem <https://edstem.org/au/courses/33564/discussion/3127162>

Warning

This practical **cannot** be completed on Windows. The Docker provider for Terraform encounters an error when creating an image on Windows. You will need to use WSL instead of the native Windows environment.

1 This Week

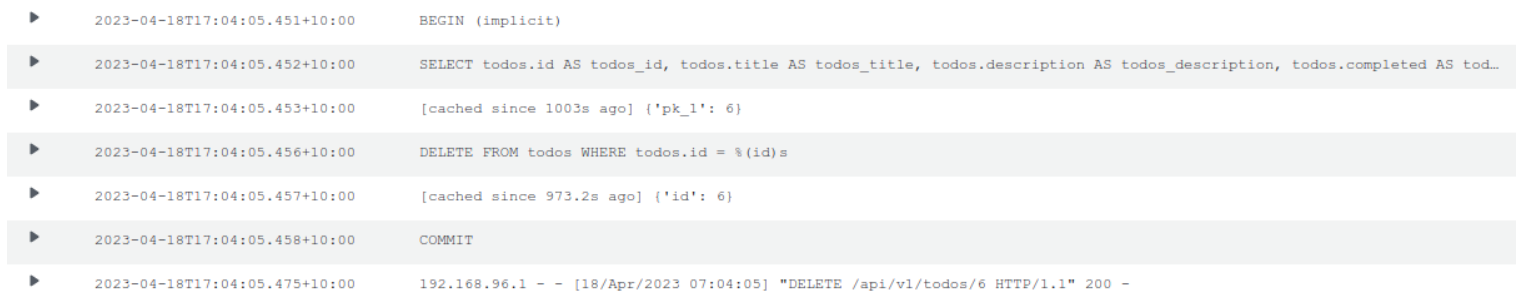
Our goal is to:

- Introduce structured logging to TaskOverflow.
- Re-deploy TaskOverflow to AWS.
- Write k6 tests to ensure TaskOverflow can handle given scenarios.
- Uncover and fix bottlenecks in the TaskOverflow application using the logs.

2 Watchtower

In this course we have repeatedly claimed that there is value in monitoring and logging. This week, we aim to prove it by using logging to help diagnose issues with a deployment of TaskOverflow. For this task, we have integrated [Watchtower](https://pypi.org/project/watchtower/)¹ into the project. Watchtower is a Python library that allows us to send logs to AWS CloudWatch, allowing us to monitor an application's performance in real time.

Currently the project is configured to log API accesses made with Flask and to log database queries created by SQL Alchemy. This results in an unstructured log stream as seen in Figure 1.



▶	2023-04-18T17:04:05.451+10:00	BEGIN (implicit)
▶	2023-04-18T17:04:05.452+10:00	SELECT todos.id AS todos_id, todos.title AS todos_title, todos.description AS todos_description, todos.completed AS tod...
▶	2023-04-18T17:04:05.453+10:00	[cached since 1003s ago] {'pk_1': 6}
▶	2023-04-18T17:04:05.456+10:00	DELETE FROM todos WHERE todos.id = %(id)s
▶	2023-04-18T17:04:05.457+10:00	[cached since 973.2s ago] {'id': 6}
▶	2023-04-18T17:04:05.458+10:00	COMMIT
▶	2023-04-18T17:04:05.475+10:00	192.168.96.1 - - [18/Apr/2023 07:04:05] "DELETE /api/v1/todos/6 HTTP/1.1" 200 -

Figure 1: An example of logs made to AWS CloudWatch for a DELETE request in the TaskOverflow API.

Getting Started

1. Using the GitHub Classroom link for this practical, provided on edstem, create a repository to work in.
2. Start your Learner Lab and copy the AWS Learner Lab credentials into a credentials file in the root of the repository.

What's New We are returning to TaskOverflow roughly from the state at the end of the *Scaling Stateless Components* practical [1]. The following notable changes have been made:

- Watchtower has been installed as a dependency.
- In `docker-compose.yml`, we mount the `credentials` file to `/root/.aws/credentials`. This allows local testing of watchtower to log to AWS CloudWatch.
- Logging has been introduced for Flask and SQL Alchemy.

¹<https://pypi.org/project/watchtower/>

Our first task will be to replicate the logs in Figure 1. Once you have copied credentials into the project root directory, start the application using docker compose.

```
$ docker compose up
```

```
app-1 | * Serving Flask app 'todo'
app-1 | * Debug mode: on
app-1 | INFO:werkzeug:WARNING: This is a development server. Do not use it in a
      | production deployment. Use a production WSGI server instead.
app-1 | * Running on all addresses (0.0.0.0)
app-1 | * Running on http://127.0.0.1:6400
app-1 | * Running on http://172.18.0.3:6400
app-1 | INFO:werkzeug:Press CTRL+C to quit
app-1 | INFO:werkzeug: * Restarting with stat
app-1 | INFO:botocore.credentials:Found credentials in shared credentials file: ~/.
      | aws/credentials
app-1 | INFO:sqlalchemy.engine.Engine:select pg_catalog.version()
app-1 | INFO:sqlalchemy.engine.Engine:[raw sql] {}
app-1 | INFO:sqlalchemy.engine.Engine:select current_schema()
app-1 | INFO:sqlalchemy.engine.Engine:[raw sql] {}
app-1 | INFO:sqlalchemy.engine.Engine:show standard_conforming_strings
app-1 | INFO:sqlalchemy.engine.Engine:[raw sql] {}
app-1 | INFO:sqlalchemy.engine.Engine:BEGIN (implicit)
app-1 | INFO:sqlalchemy.engine.Engine:SELECT pg_catalog.pg_class.relname
app-1 | FROM pg_catalog.pg_class JOIN pg_catalog.pg_namespace ON pg_catalog.
      | pg_namespace.oid = pg_catalog.pg_class.relnamespace
app-1 | WHERE pg_catalog.pg_class.relname = %(table_name)s::VARCHAR AND pg_catalog.
      | pg_class.relkind = ANY (ARRAY[% (param_1)s::VARCHAR, %(param_2)s::VARCHAR, %(
      | param_3)s::VARCHAR, %(param_4)s::VARCHAR, %(param_5)s::VARCHAR]) AND pg_catalog.
      | pg_table_is_visible(pg_catalog.pg_class.oid) AND pg_catalog.pg_namespace.nspname
      | != %(nspname_1)s::VARCHAR
app-1 | INFO:sqlalchemy.engine.Engine:[generated in 0.00015s] {'table_name': 'todos',
      | 'param_1': 'r', 'param_2': 'p', 'param_3': 'f', 'param_4': 'v', 'param_5': 'm',
      | 'nspname_1': 'pg_catalog'}
app-1 | INFO:sqlalchemy.engine.Engine:COMMIT
app-1 | WARNING:werkzeug: * Debugger is active!
app-1 | INFO:werkzeug: * Debugger PIN: 123-456-789
```

You should see logs similar to the above. Notice that information about Flask is prefixed with `INFO:werkzeug` and information about SQL Alchemy is prefixed with `INFO:sqlalchemy.engine.Engine`. These prefixes indicate into which stream the logs are put.

Open the AWS CloudWatch console and go to `Logs > Log Management` on the side panel. You should see a log group called `taskoverflow`. If you click on that group you will see log streams associated with `INFO:werkzeug` and `INFO:sqlalchemy.engine.Engine`.

The screenshot shows the AWS CloudWatch console interface. On the left is a navigation menu with sections like 'CloudWatch', 'Log Management', 'AI Operations', 'GenAI Observability', 'Application Signals (APM)', 'Infrastructure Monitoring', 'Logs', and 'Metrics'. The main content area is titled 'taskoverflow' and contains 'Log group details' and 'Log streams (4)'. The 'Log group details' section shows the log class as 'Standard', ARN as 'arn:aws:logs:us-east-1:163245180706:log-group:taskoverflow:*', and other configuration options like retention and deletion protection. The 'Log streams (4)' section displays a table of log streams with their names and last event times.

Log stream	Last event time
d389fc43509c//root/.cache/pypoetry/virtualenvs/practical08-9TtSrW0h-py3.12/bin/flask/werkzeug/19	2026-04-17 12:02:45 (UTC)
d389fc43509c//root/.cache/pypoetry/virtualenvs/practical08-9TtSrW0h-py3.12/bin/flask/sqlalchemy.engine.Engine	2026-04-17 12:02:45 (UTC)
d389fc43509c//root/.cache/pypoetry/virtualenvs/practical08-9TtSrW0h-py3.12/bin/flask/werkzeug/1	2026-04-17 12:02:44 (UTC)
d389fc43509c//root/.cache/pypoetry/virtualenvs/practical08-9TtSrW0h-py3.12/bin/flask/sqlalchemy.engine.Engine	2026-04-17 12:02:44 (UTC)

Figure 2: TaskOverflow log streams in CloudWatch.

2.1 Structured Logging

Our first task will be to convert the current logging into a structured logging format. As we saw in the logging lecture [2], structured logging can be as simple as logging a JSON object. This allows logging services to quickly filter through logs based on criteria selected from the object's fields.

In `todo/__init__.py` we have the following code within the `create_app` function. This code configures watchtower to log to AWS CloudWatch.

```
» cat todo/__init__.py

def create_app(config_overrides=None):
    ...
    handler = watchtower.CloudWatchLogHandler(
        log_group_name="taskoverflow",
        boto3_client=boto3.client("logs", region_name="us-east-1")
    )
    app.logger.addHandler(handler)
    ...
```

We want to wrap all our logs in JSON objects and inject metadata into these objects. This makes it easier to search the logs. To do this, we will create a custom log formatter that is a subclass of `watchtower.CloudWatchLogFormatter`.

```
» cat todo/log_formatter.py
```

```

import logging
import watchtower
from flask import has_request_context, request

class StructuredFormatter(watchtower.CloudWatchLogFormatter):
    def format(self, record):
        record.msg = {
            'timestamp': record.created,
            'location': record.name,
            'message': record.msg,
        }
        if has_request_context():
            record.msg['request_id'] = request.environ.get('REQUEST_ID')
            record.msg['url'] = request.environ.get('PATH_INFO')
            record.msg['method'] = request.environ.get('REQUEST_METHOD')
        return super().format(record)

```

These are just some metadata fields we might want to add to our logging. We may also consider `socket.gethostname()` to identify the instance handling a request, or other helpful information.

We then need to get the `CloudWatchLogHandler` to use our `StructuredFormatter`. Add the following code to the `todo/__init__.py` file.

```

» cat todo/__init__.py

from .log_formatter import StructuredFormatter
...
def create_app(config_overrides=None):
    ...
    handler = ...
    handler.setFormatter(StructuredFormatter())
    ...

```

2.2 Correlation IDs

Correlation IDs are a mechanism to help understand the path of a request, event, message, etc. through a system [2]. When logging it is often helpful to be able to trace the execution of a particular request.

For our system we will generate a new random identifier for each incoming request. This identifier will be included as part of the logging metadata. A system with multiple API endpoints requests, will often come with an established `REQUEST_ID` header.

```

» cat todo/__init__.py

import uuid
from flask import Flask, request
...

def create_app(config_overrides=None):
    ...
    requests = logging.getLogger("requests")

```

```
requests.addHandler(handler)

@app.before_request
def before_request():
    request.environ['REQUEST_ID'] = str(uuid.uuid4())
    requests.info("Request started")

@app.after_request
def after_request(response):
    requests.info("Request finished")
    return response

# Load the models
...
```

The code above in `create_app` will generate a unique identifier for each incoming request. It will also log when a request starts being processed and when it finishes processing.

2.3 Verify and Refine

Ensure that the above modifications are working as expected by returning to the AWS CloudWatch logs. Launch the service locally and make some API requests.

You should now be able to go to the `Log Insights` interface from the sidebar. This interface allows you to query structured logging data. Select the `taskoverflow` log group and run the following query to ensure that your logs seem sensible.

```
fields request_id, url, method, @timestamp, message.message, @message, @method
| filter not isempty(request_id)
| limit 400
```

Take some time now to refine your logging to produce logs that are easy to search through and to understand how to trace API requests.

3 Load Testing

We will now generate some load on our API using `k6` and see how it performs. We briefly saw `k6` at the end of the *Scaling Stateless Components* practical [1]. To review, `k6` is a load testing tool that is written in Go but provides an interface using a subset of JavaScript.

It will help us to generate a large number of concurrent API requests. If you have not already done so, install `k6` from <https://k6.io/docs/get-started/installation/>.

3.1 Scenario

We want to simulate the following scenario using our testing framework.

Indecisive planners and studious reviewers Our TaskOverflow application is being used at a point in semester where most students have already setup their tasks. These students are routinely visiting the website and listing the tasks they have yet to complete. At the same time, there are 40 very indecisive students who, quite late in semester, are trying to setup their tasks for the rest of semester. They continually create a task, realise that they have mis-typed, delete the task, and start over.

As the tasks are shared globally amongst all students, the indecisive planners are altering the list of task seen by the organised students.

3.2 Setup

To get started, we will create a file called `planners-and-studiers.js`. Most k6 tests start with the following imports.

```
» cat planners-and-studiers.js

import http from "k6/http";
import { check, sleep } from "k6";

const ENDPOINT = __ENV.ENDPOINT;
```

- `http` holds the methods used to make HTTP requests,
- `check` allows us to assert the state of HTTP responses, and
- `sleep` gives us the ability to put a simulated user to sleep rather than continuously spamming the service with requests.

The `ENDPOINT` constant retrieves the endpoint URL from the `ENDPOINT` environment variable.

3.3 User Simulation

The behaviour of our users will be defined by an exported JavaScript function. Our studying student will be listing out all of the tasks they have left to complete by using the `/api/v1/todos` endpoint.

```
» cat planners-and-studiers.js

export function studyingStudent() {
  let url = ENDPOINT + '/api/v1/todos';

  // What tasks do I have left to work on?
  let request = http.get(url);

  check(request, {
    'is status 200': (r) => r.status === 200,
  });

  // Alright I'll go work on my next task for around 2 minutes
  sleep(120);
}
```


Of course, this test is very basic as it only ensures the response code is 200. There is no guarantee that the returned data is sensible.

Info

If you would like a challenge, you can use the `randomIntBetween` function to have the studier occasionally tick off tasks.

<https://k6.io/docs/javascript-api/jslib/utils/randomintbetween/>

Next we need to simulate our indecisive students. For them, we will need to make a POST request to the `/api/v1/todos` endpoint to create a task. Then we will need to use the ID given by the POST response to DELETE the mis-typed task.

```
» cat planners-and-studiers.js

export function indecisivePlanner() {
  let url = ENDPOINT + '/api/v1/todos';

  // I need to work on the CSSE6400 Cloud Assignment!
  const payload = JSON.stringify({
    "title": "CSSE6400 Clout Assignment",
    "completed": false,
    "description": "",
    "deadline_at": "2026-05-01T15:00:00",
  });

  const params = {
    headers: {
      'Content-Type': 'application/json',
    },
  };

  let request = http.post(url, payload, params);
  check(request, {
    'is status 200': (r) => r.status === 200,
  });

  sleep(10);

  // Oh no! Not the Clout assignment, the Cloud assignment!
  const wrongId = request.id;

  request = http.del(`${url}/${wrongId}`);

  check(request, {
    'is status 200': (r) => r.status === 200,
  });

  // I'll come back to it later :(
  sleep(10);
}
```


3.4 Configure Behaviour

We have outlined how our user agents will interact with our API. The `options` object allows us to configure how these interactions occur over time. To prevent overloading the API immediately, and allow any auto-scaling behaviour to be triggered, we will gradually increase the amount of studiers. We will have a consistent amount of planners, 20, who will perform a total of 400 corrections over the test run.

```
» cat planners-and-studiers.js
export const options = {
  scenarios: {
    studier: {
      exec: 'studyingStudent',
      executor: 'ramping-vus',
      stages: [
        { duration: '1m', target: 1500 },
        { duration: '3m', target: 7500 },
        { duration: '2m', target: 0 },
      ],
    },
    planner: {
      exec: 'indecisivePlanner',
      executor: 'shared-iterations',
      vus: 20,
      iterations: 400,
    },
  },
};
```

3.5 Running the Tests

Now that we have implemented a load test. Deploy the service to AWS using Terraform and test it.

```
$ terraform init
```

```
$ terraform apply
```

Once the deployment is finished, Terraform should return the URL where the service is deployed. Query the health endpoint to make sure the service is running. Use the URL to set the `ENDPOINT` environment variable and run the tests.

```
> export ENDPOINT=...
> k6 run planners-and-studiers.js
```

These tests should take less than 10 minutes to run. Most of the tests should pass but there should be a few failures throughout, as seen below. Note that the exact number and percentages of successes and failures will be unique to each test run.

```

WARN[0152] Request Failed error="Get \"http://taskoverflow-451078826.us-east-1.elb.
amazonaws.com/api/v1/todos\": dial: i/o timeout"
WARN[0183] Request Failed error="Get \"http://taskoverflow-451078826.us-east-1.elb.
amazonaws.com/api/v1/todos\": dial: i/o timeout"
WARN[0239] Request Failed error="Get \"http://taskoverflow-451078826.us-east-1.elb.
amazonaws.com/api/v1/todos\": dial: i/o timeout"
WARN[0241] Request Failed error="Delete \"$%7Burl%7D/%$%7BwrongId%7D\": unsupported
protocol scheme \"\""
WARN[0272] Request Failed error="Get \"http://taskoverflow-451078826.us-east-1.elb.
amazonaws.com/api/v1/todos\": dial: i/o timeout"
WARN[0330] Request Failed error="Get \"http://taskoverflow-451078826.us-east-1.elb.
amazonaws.com/api/v1/todos\": dial: i/o timeout"
WARN[0339] Request Failed error="Get \"http://taskoverflow-451078826.us-east-1.elb.
amazonaws.com/api/v1/todos\": dial: i/o timeout"
WARN[0371] Request Failed error="request timeout"
WARN[0412] Request Failed error="Get \"http://taskoverflow-451078826.us-east-1.elb.
amazonaws.com/api/v1/todos\": dial: i/o timeout"

```

TOTAL RESULTS

```

checks_total.....: 28105 70.103828/s
checks_succeeded.....: 92.47% 25989 out of 28105
checks_failed.....: 7.52% 2116 out of 28105

```

```

is status 200
92% - 15257 / 1241

```

HTTP

```

http_req_duration.....: avg=7.09s min=0s med
=964.15ms max=1m1s p(90)=21.72s p(95)=27.89s
{ expected_response:true }.....: avg=7.19s min=214.35
ms med=1s max=1m0s p(90)=22.01s p(95)=28.39s
http_req_failed.....: 3.88% 1092 out of
28105
http_reqs.....: 28105 70.103828/s

```

EXECUTION

```

iteration_duration.....: avg=2m0s min=20.21s
med=2m0s max=2m27s p(90)=2m8s p(95)=2m12s
iterations.....: 15492 38.642537/s
vus.....: 27 min=27 max=15020
vus_max.....: 15020 min=15020 max
=15020

```

NETWORK

```

data_received.....: 969 MB 2.4 MB/s
data_sent.....: 3.3 MB 8.3 kB/s

```

```
running (07m22.2s), 0000/15020 VUs, 15492 complete and 12377 interrupted iterations
planner [=====] 20 VUs 07m22.2s/10m0s 400/400 shared
    iters
studier [=====] 0000/15000 VUs 6m0s
```

4 Debugging & Scaling Improvements

Explore the logs to determine if there are any errors that are causing some of the failures. Consider how to improve the service's ability to scale. Increase the number of studiers and planners and use the logs to look for bottlenecks. Design other scenarios to explore other usage patterns. We want to increase our 92% pass rate to be closer to 100%.

In almost all systems, it is possible to design test scenarios that will have timeout failures. The goal is usually not to have a system that can handle any load, but to design a system that can handle expected load behaviours and have some overhead to handle reasonable unexpected load.

4.1 Tips

Open TaskOverflow If you open TaskOverflow, you should find it has been left in an unusual state after the tests. See if you can find out from the logs how this occurred.

Follow an Interaction Use the correlation IDs that we have included to see where potential bottlenecks may be occurring. You can introduce more fine-grained logs to identify where time is being spent.

References

- [1] M. Hughes, A. Truffet, B. Webb, and R. Thomas, "Scaling stateless components," vol. 6 of *CSSE6400 Practicals*, The University of Queensland, April 2025. <https://csse6400.uqcloud.net/practicals/week06.pdf>.
- [2] L. Le Gassick, R. Thomas, and B. Guandong, "Monitoring and logging slides," April 2026. https://docs.google.com/presentation/d/1WCPphBCVq_KgKQtvD4ys-QuKq2rA49JHVWCZ-FUAEFg/edit?usp=sharing.